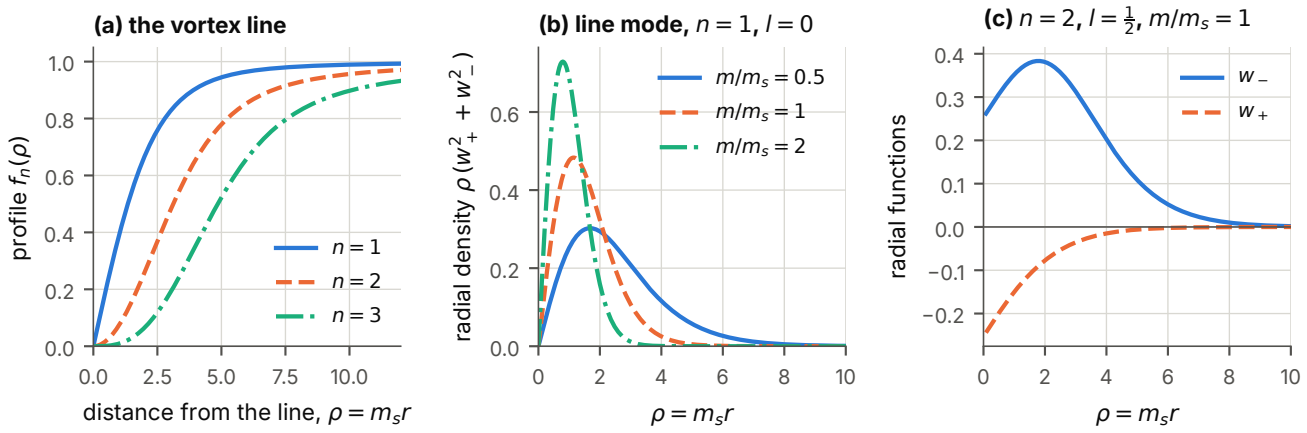
(a) the vortex line
profile $f_n(\rho)$
distance from the line, $\rho = m_s r$
$n = 1$
$n = 2$
$n = 3$
(b) line mode, $n = 1$, $l = 0$
radial density $\rho\,(w_+^2 + w_-^2)$
$\rho = m_s r$
$m/m_s = 0.5$
$m/m_s = 1$
$m/m_s = 2$
(c) $n = 2$, $l = \frac{1}{2}$, $m/m_s = 1$
radial functions
$\rho = m_s r$
w_-
w_+